Compiler Construction

Lab 5

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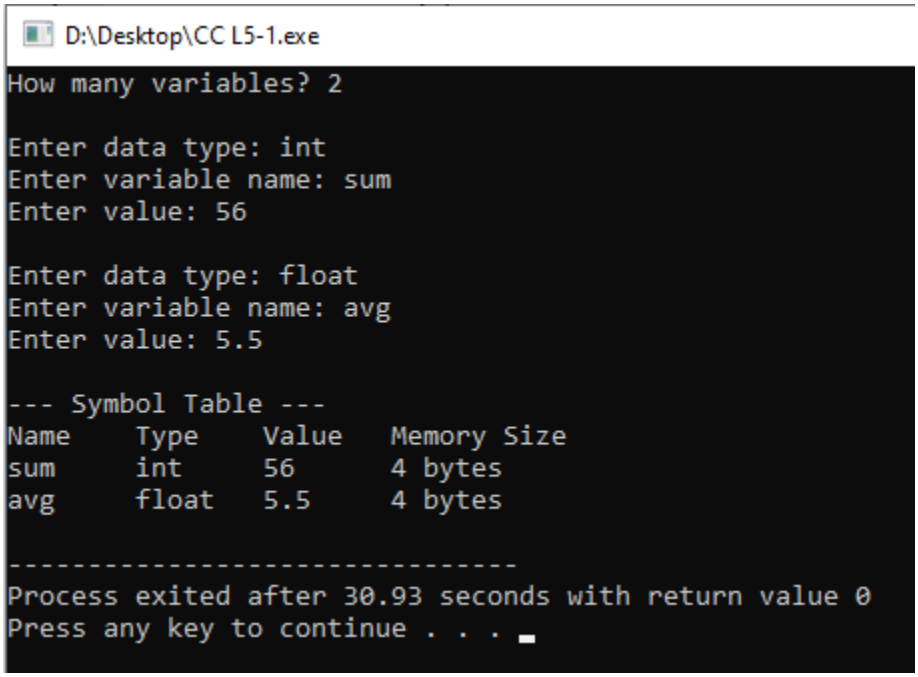
Date: February 24th, 2026

Task 1:

```
#include <iostream>
using namespace std;
struct Symbol
{
    string name;
    string type;
    string value;
    int memory_size;
};
int main()
{
    Symbol table[10];
    int n, count = 0;
    cout << "How many variables? ";
    cin >> n;
    for(int i = 0; i < n; i++)
    {
        string name, type, value;
        bool duplicate = false;
        cout << "\nEnter data type: ";
        cin >> type;
        cout << "Enter variable name: ";
        cin >> name;
        cout << "Enter value: ";
        cin >> value;
        for(int j = 0; j < count; j++)
        {
```

```
        if(table[j].name == name)
        {
            cout << "Error: Duplicate identifier!\n";
            duplicate = true;
            break;
        }
    }
    if(!duplicate)
    {
        table[count].name = name;
        table[count].type = type;
        if(table[count].type=="int")
        {
                table[count].memory_size=4;
                        }
                        else if(table[count].type=="float")
        {
                table[count].memory_size=4;
                        }
                        else if(table[count].type=="char")
        {
                table[count].memory_size=1;
                        }
        table[count].value = value;
        count++;
    }
}
```

```
    cout << "\n--- Symbol Table ---\n";
    cout << "Name\tType\tValue\tMemory Size\n";
    for(int i = 0; i < count; i++)
    {
        cout << table[i].name << "\t"
            << table[i].type << "\t"
            << table[i].value << "\t"
            << table[i].memory_size << " bytes" << endl;
    }
    return 0;
}
```



Task 2:

```
#include <iostream>
using namespace std;
struct Symbol
{
    string name;
    string type;
    string value;
    int memory_size;
};
int main()
{
    Symbol table[10];
    int n, count = 0;
    cout << "How many variables? ";
    cin >> n;
    for(int i = 0; i < n; i++)
    {
        string name, type, value;
        bool duplicate = false;
        cout << "\nEnter data type: ";
        cin >> type;

        cout << "Enter variable name: ";
        cin >> name;
        cout << "Enter value: ";
        cin >> value;
        for(int j = 0; j < count; j++)
```

```
    {
        if(table[j].name == name)
        {
            cout << "Error: Duplicate identifier!\n";
            duplicate = true;
            break;
        }
    }
    if(!duplicate)
    {
        table[count].name = name;
        table[count].type = type;
        if(table[count].type=="int")
        {
                table[count].memory_size=4;
                        }
                        else if(table[count].type=="float")
        {
                table[count].memory_size=4;
                        }
                        else if(table[count].type=="char")
        {
                table[count].memory_size=1;
                        }
        table[count].value = value;
        count++;
    }
}
```

```
cout << "\n--- Symbol Table ---\n";
cout << "Name\tType\tValue\tMemory Size\n";
for(int i = 0; i < count; i++)
{
   cout << table[i].name << "\t"
        << table[i].type << "\t"
        << table[i].value << "\t"
        << table[i].memory_size << " bytes" << endl;
}
string identifier;
bool found;
       cout << "\nEnter indentifier for lookup operation: ";
       cin >> identifier;

       for(int i = 0; i < n; i++)
{
       if(table[i].name==identifier)
       {
                cout << "\n--- Symbol Table ---\n";
                cout << "Name\tType\tValue\tMemory Size\n";
                cout << table[i].name << "\t"
        << table[i].type << "\t"
        << table[i].value << "\t"
        << table[i].memory_size << " bytes" << endl;
       cout << "Identifier Found!";
       found=true;
       break;
                }
```

```
}
        if(!found)
        cout << "Identifier Not Found!";
    return 0;
}
```

```
D:\Desktop\CC L5-2.exe
How many variables? 3

Enter data type: int
Enter variable name: a
Enter value: 11

Enter data type: float
Enter variable name: b
Enter value: 1.4

Enter data type: char
Enter variable name: c
Enter value: A

--- Symbol Table ---
Name    Type    Value   Memory Size
a       int     11      4 bytes
b       float   1.4     4 bytes
c       char    A       1 bytes

Enter indentifier for lookup operation: c

--- Symbol Table ---
Name    Type    Value   Memory Size
c       char    A       1 bytes
Identifier Found!
--------------------------------
Process exited after 52.02 seconds with return value 0
Press any key to continue . . .
```

Task 3:

```
#include <iostream>
using namespace std;

struct Symbol
{
    string name;
    string type;
    string value;
};

int main()
{
    Symbol table[10];
    int n, count = 0;

    cout << "How many variables? ";
    cin >> n;

    for(int i = 0; i < n; i++)
    {
        string name, type, value;
        bool duplicate = false;

        cout << "\nEnter data type: ";
        cin >> type;

        cout << "Enter variable name: ";
```

```
    cin >> name;

    cout << "Enter value: ";
    cin >> value;

    for(int j = 0; j < count; j++)
    {
        if(table[j].name == name)
        {
            cout << "Error: Duplicate identifier!\n";
            duplicate = true;
            break;
        }
    }

    if(!duplicate)
    {
        table[count].name = name;
        table[count].type = type;
        table[count].value = value;
        count++;
    }
}

cout << "\n--- Symbol Table ---\n";
cout << "Name\tType\tValue\n";

for(int i = 0; i < count; i++)
```

```
    {
        cout << table[i].name << "\t"
            << table[i].type << "\t"
            << table[i].value << endl;
    }

    return 0;
}
```

```
D:\Desktop\CC L5-3.exe
How many variables? 2

Enter data type: string
Enter variable name: name
Enter value: ayesha

Enter data type: int
Enter variable name: name
Enter value: 10
Error: Duplicate identifier!

--- Symbol Table ---
Name    Type    Value
name    string  ayesha

--------------------------------
Process exited after 26.71 seconds with return value 0
Press any key to continue . . .
```